

Here stand two *similar* figures: the first figure ABC and the second figure DEF . To say that two figures are *similar* means that they have exactly the same shape, though not necessarily the same size — one may be a scaled copy of the other. All their corresponding angles are equal ($\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$), and their corresponding sides run in the same proportion throughout. The matching outline shared by the larger figure and the smaller figure — their common shape — is the heart of this lemma. Whatever we now prove about one such pair holds for *all* similar figures, whether bounded by straight lines or by curves.